

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 5

Duration: 1 Hour

Total Marks:50

Annexure A

Analytical Problem Solving

1. A coin is tossed 10 times and the outcomes are:

H H T H H H T H H T

Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).

Answers:

1. Coin Toss (MLE for Probability of Heads)

Given:

Outcomes: **H H T H H H T H H T**

- Total tosses (**n**) = 10
- Number of Heads (**H**) = 7

Formula:

MLE = Number of Heads / Total Tosses

Calculation:

$$\hat{p} = 7 / 10 = 0.7$$

Answer: The MLE of the probability of getting Heads is **0.7 (70%)**.

2. Out of 50 students, 42 passed an exam.

Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.

Answers:

2. Students Passing an Exam

Given:

- Total students = 50
- Students passed = 42

Formula:

MLE = Number of Successes / Total Observations

Calculation:

$$\hat{p} = 42 / 50 = 0.84$$

Answer: The MLE of the probability that a student pass is **0.84**.

3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective. Find the MLE for the probability that a bulb is defective.

Answers:

3. Defective Light Bulbs

Given:

- Total bulbs = 100
- Defective bulbs = 8

Formula:

MLE = Number of Defective Bulbs / Total Bulbs

Calculation:

$$\hat{p} = 8 / 100 = 0.08$$

Answer: The MLE of the probability that a bulb is defective is **0.08**

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4. A biased die is rolled 60 times. The number 6 appears 18 times. Estimate the probability of rolling a 6 using MLE.

Answers:

4. Biased Die

Given:

- Total rolls = 60
- Number of times 6 appears = 18

Formula:

MLE = Number of Successes / Total Trials

Calculation:

$$\hat{p} = 18 / 60 = 0.30$$

Answer: The MLE of the probability of rolling a 6 is 0.30

5. Given:

Input (x) = 4

Weight (w) = 2

Actual Output = 10

Prediction:

$$y = w \times x$$

1. Find the predicted output.
2. Calculate the error.
3. If the gradient is 4 and learning rate is 0.01, find the updated weight.

Answers:

5. Prediction, Error, and Weight Update

Given:

- Input (x) = 4
- Weight (w) = 2
- Actual Output = 10
- Learning Rate (α) = 0.01
- Gradient = 4

1. Predicted Output

Formula:

$$y = w \times x$$

Calculation:

$$y = 2 \times 4 = 8$$

Answer: Predicted Output = 8

2. Error

Formula:

Error = Actual Output – Predicted Output

Calculation:

$$\text{Error} = 10 - 8 = 2$$

Answer: Error = 2

3. Updated Weight

Formula:

New Weight = Old Weight – (Learning Rate $\times$ Gradient)

Calculation:

$$\text{New Weight} = 2 - (0.01 \times 4)$$

$$\text{New Weight} = 2 - 0.04 = 1.96$$

Answer: Updated Weight = 1.96

Code of Conduct:

I , M Sohail Khan(192525046)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

M Sohail Khan

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and	Understands problem with minor gaps	Partial understanding;	Misinterprets or unable to

		requirements accurately		misses key elements	understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation